

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course:** B.Tech **Branch :** Electrical Engineering and Allied**Semester :** VI**Subject Code & Name:** BTEEPE604B-Smart Grid Technology**Max Marks:** 60**Date:** 26/12/2024**Duration:** 3 Hrs.**Instructions to the Students:**

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 9.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculator is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	One Nation, One Grid, One _____.	K1/CO1	1
	a. Current b. Frequency c. Resistances d. Voltage		
2	Smart grids rely on for real-time data collection and communication.	K1/CO2	1
	A. Outage Management System B. Smart Meters C. Scada D. Artificial Intelligence		
3	What is a significant advantage of microgrids?	K1/CO3	1
	a. High dependency on fossil fuels b. Increased risk of power outages c. Ability to operate in island mode during main grid failures d. Limited access to renewable energy integration		
4	Which of the following is NOT a benefit of electric vehicles?	K1/CO3	1
	a. Reduced greenhouse gas emissions b. Lower operating costs c. Faster refueling than gasoline vehicles d. Decreased reliance on fossil fuels		
5	Which of the following communication technologies is most suitable for low-power IoT applications?	K1/CO4	1
	a. ZigBee b. GPS c. Wi-Fi d. WAN		
6	In IP-based systems, what does IP stand for?	K1/CO4	1
	a. International Protocol b. Internet Protocol c. Interpersonal Protocol d. Integrated Process		
7	What is the primary objective of Load Frequency Control (LFC) in microgrids?	K1/CO5	1
	a. Maintain constant voltage levels b. Stabilize the frequency of the system c. Increase system losses d. Enhance power transmission efficiency		
8	Which of the following is an example of a smart grid testbed?	K1/CO5	1
	a. National Renewable Energy Lab (NREL) Testbed b. NASA Communication Testbed c. CERN Particle Accelerator Testbed d. Global Automotive Grid Testbed		
9	Renewable energy is generated from.....?	K1/CO2	1
	a. Natural resources B. artificial Resources c. Carbon resources d. Nuclear resources		

10	Compressed Air Energy Storage (CAES) utilities us.....?				K1/CO1	1
	a. electricity generated during off - peak hours to compress air	b. electricity generated during peak hours to compress air	c. does not need electricity to compress air	d. use air at atmospheric pressure only		
11	Which of the following is NOT a type of smart energy storage?				K1/CO1	1
	a. Pumped Hydro Storage	b. Compressed Air Energy Storage	c. Lead-acid batteries in traditional cars	d. superconducting Magnetic Energy Storage (SMES)		
12	What is a key feature of Smart Grid architecture?				K1/CO2	1
	a. One-way communication	b. Two-way communication between utilities and consumers	c. Centralized control without monitoring	d. Manual data collection		
Q. 2 Solve the following.						12
A)	Describe the Intelligent Electronic Devices (IED) with their application				K2/CO1	6
B)	Illustrate the Compressed Air used for renewable energy smart storage.				K3/CO1	6
Q.3 Solve the following.						12
A)	Describe Architecture of Smart Grid with suitable layout diagram.				K2/CO2	6
B)	Discuss specific challenges associated with integrating renewable energy sources into the Smart Grid?				K2/CO2	6
Q. 4 Solve Any Two of the following.						12
A)	Explain working of Microgrid with its benefits and drawbacks.				K2/CO3	6
B)	Discuss the Environmental impact and Climate Change due to electrical power generation.				K2/CO3	6
C)	Describe economic issues come under micro grids.				K2/CO3	6
Q.5 Solve Any Two of the following.						12
A)	Explain the role of IP-based systems and Power Line Communications (PLC) in smart grid communication networks.				K2/CO4	6
B)	Discuss the concept of a two-way digital communications paradigm in the context of smart grids with benefits.				K2/CO4	6
C)	Illustrate Principle of Phasor Measurement Units (PMUs) with suitable diagram.				K2/CO4	6
Q. 6 Solve Any Two of the following.						12
A)	Explain the Load Frequency Control (LFC) and Voltage Control in Micro Grid System				K2/CO5	6
B)	Discuss any one case study related to smart grid implementation				K2/CO5	6
C)	Illustrate the biggest cybersecurity threats facing smart grids, and how can these threats be mitigated?				K3/CO5	6
*** End ***						